INFOSYS SE PSEUDOCODE VAULT • BITWISE OPERATORS & LOGIC

XOR, AND, OR, Bit Shifts, Precedence, Compound Expressions • Total Solved Questions: 93

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Operator Precedence

Q1. What will be the output of the following pseudocode?

```
Set Integer a = 18
Set Integer b = 5
Set Integer c = 2
display a - b * c + 6
```

(A) 10
(B) 14
(C) 16
(D) 19

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** Multiplication first: 18 – (5×2) + 6 = 14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Arithmetic

Q2. What will be the output of the following pseudocode?

```
Set Integer x = 24
Set Integer y = 7
display (x / y) * y + (x % y)
```

(A) 21
(B) 24
(C) 25
(D) 28

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** 24/7=3 and 24%7=3; 3×7+3 = 24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Logical Operators

Q3. What will be the output of the following pseudocode?

```
Set Integer a = 12
Set Integer b = 7
Set Integer c = 5
display (a > b) and ((b - c) == 2)
```

(A) true
(B) false
(C) 2
(D) 12

Correct Answer: Option (A) : true

■ **Step-by-Step Logic:** 12>7 is true and 7–5=2 is true, so the result is true.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Logical Operators

Q4. What will be the output of the following pseudocode?

```
Set Integer x = 9
display (x % 2 == 0) or ((x > 5) and (x < 10))
```

(A) true
(B) false
(C) 9
(D) 0

Correct Answer: Option (A) : true

■ **Step-by-Step Logic:** First condition is false, but 9>5 and 9<10 are both true.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Bitwise XOR

Q5. What will be the output of the following pseudocode?

```
Set Integer a = 15
Set Integer b = 9
Set Integer c = 6
display (a ^ b) + c
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (12).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Bitwise AND

Q6. What will be the output of the following pseudocode?

```
Set Integer a = 27
Set Integer b = 14
display (a & b) + (a % b)
```

(A) 19
(B) 21
(C) 23
(D) 25

Correct Answer: Option (C) : 23

■ **Step-by-Step Logic:** 27&14 = 10 and 27%14 = 13; total = 23.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Bitwise OR

Q7. What will be the output of the following pseudocode?

```
Set Integer a = 18
Set Integer b = 11
display (a | b) - (a & b)
```

(A) 21
(B) 23
(C) 25
(D) 27

Correct Answer: Option (C) : 25

■ **Step-by-Step Logic:** 18|11=27 and 18&11=2; 27–2=25.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: XOR + AND

Q8. What will be the output of the following pseudocode?

```
Set Integer a = 13
Set Integer b = 7
display (a ^ b) + (a & b)
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** 13^7=10 and 13&7=5; total = 15. Infosys Medium Pseudocode Practice • Page 33

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Pre/Post Increment

Q9. What will be the output of the following pseudocode?

```
Set Integer x = 4
Set Integer y = x++ + ++x
Set Integer z = y + x
display z
```

(A) 14
(B) 15
(C) 16
(D) 17

Correct Answer: Option (C) : 16

■ **Step-by-Step Logic:** x++ contributes 4 and makes x=5; ++x makes x=6 and contributes 6. y=10; z=10+6=16.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Post/Pre Decrement

Q10. What will be the output of the following pseudocode?

```
Set Integer x = 10
Set Integer y = --x
Set Integer z = y + x--
display z
```

(A) 16
(B) 17
(C) 18
(D) 19

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** --x makes x=9, y=9; x-- contributes 9, then x becomes 8. z=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Relational Logic

Q11. What will be the output of the following pseudocode?

```
Set Integer p = 8
Set Integer q = 12
display (p + q > 20) and (q - p == 4)
```

(A) true
(B) false
(C) 4
(D) 20

Correct Answer: Option (B) : false

■ **Step-by-Step Logic:** p+q=20, so 20>20 is false. The second condition is true; false AND true = false.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Ternary Logic

Q12. What will be the output of the following pseudocode?

```
Set Integer a = 14
Set Integer b = 9
Set Integer m = (a % 2 == 0) ? a/2 : b*2
display m + b
```

(A) 14
(B) 16
(C) 18
(D) 21

Correct Answer: Option (B) : 16

■ **Step-by-Step Logic:** a is even, so m=7; 7+9=16.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Compound Assignment

Q13. What will be the output of the following pseudocode?

```
Set Integer x = 7
x = x * 3 - 4
x = x + (x % 5)
display x
```

(A) 18
(B) 19
(C) 20
(D) 21

Correct Answer: Option (B) : 19

■ **Step-by-Step Logic:** 7×3–4=17; 17%5=2; final x=19.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Operators

Q14. What will be the output of the following pseudocode?

```
Set Integer a = 20
Set Integer b = 6
display (a / b) + (a % b) * 2
```

(A) 5
(B) 7
(C) 9
(D) 11

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** 20/6=3; 20%6=2; 3+2×2=7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Left Shift

Q15. What will be the output of the following pseudocode?

```
Set Integer x = 11
Set Integer y = 2
display (x << y) - x
```

(A) 22
(B) 33
(C) 44
(D) 55

Correct Answer: Option (B) : 33

■ **Step-by-Step Logic:** 11<<2 = 44; 44–11=33.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Right Shift

Q16. What will be the output of the following pseudocode?

```
Set Integer x = 45
Set Integer y = 2
display (x >> y) + x % 4
```

(A) 10
(B) 11
(C) 12
(D) 13

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** 45>>2 = 11; 45%4=1; total=12. Infosys Medium Pseudocode Practice • Page 34

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Left Shift in Expression

Q17. What will be the output of the following pseudocode?

```
Set Integer a = 6
Set Integer b = 3
display (a << b) / 4
```

(A) 8
(B) 10
(C) 12
(D) 16

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (12).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Right Shift in Expression

Q18. What will be the output of the following pseudocode?

```
Set Integer a = 50
display (a >> 1) - (a >> 3)
```

(A) 17
(B) 19
(C) 21
(D) 23

Correct Answer: Option (B) : 19

■ **Step-by-Step Logic:** 50>>1=25 and 50>>3=6; 25–6=19.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + XOR

Q19. What will be the output of the following pseudocode?

```
Set Integer x = 9
display (x << 1) ^ x
```

(A) 19
(B) 23
(C) 27
(D) 31

Correct Answer: Option (C) : 27

■ **Step-by-Step Logic:** 9<<1=18; 18 XOR 9 = 27.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + AND

Q20. What will be the output of the following pseudocode?

```
Set Integer x = 29
display (x >> 2) & 7
```

(A) 5
(B) 6
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** 29>>2=7; 7 AND 7 = 7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Double Shift

Q21. What will be the output of the following pseudocode?

```
Set Integer x = 7
Set Integer y = (x << 2)
display y >> 1
```

(A) 7
(B) 12
(C) 14
(D) 28

Correct Answer: Option (C) : 14

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (14).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift and Modulo

Q22. What will be the output of the following pseudocode?

```
Set Integer x = 37
display (x >> 1) % 6
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (A) : 0

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (0).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift Condition

Q23. What will be the output of the following pseudocode?

```
Set Integer x = 23
if ((x >> 2) >= 5)
display x - 3
else
display x + 3
end-if
```

(A) 20
(B) 23
(C) 26
(D) 28

Correct Answer: Option (A) : 20

■ **Step-by-Step Logic:** 23>>2=5, so the condition is true and 23–3=20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + Arithmetic

Q24. What will be the output of the following pseudocode?

```
Set Integer a = 14
Set Integer b = 2
display (a << b) - (a >> b)
```

(A) 51
(B) 53
(C) 55
(D) 57

Correct Answer: Option (B) : 53

■ **Step-by-Step Logic:** 14<<2=56; 14>>2=3; 56–3=53. Infosys Medium Pseudocode Practice • Page 35

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Topic: Shift Chain

Q25. What will be the output of the following pseudocode?

```
Set Integer x = 3
x = x << 2
x = x >> 1
x = x << 1
display x
```

(A) 6
(B) 10
(C) 12
(D) 24

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (12).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shift + OR

Q26. What will be the output of the following pseudocode?

```
Set Integer x = 5
display (x << 2) | 3
```

(A) 19
(B) 20
(C) 23
(D) 24

Correct Answer: Option (C) : 23

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (23).

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Topic: Shift + XOR

Q27. What will be the output of the following pseudocode?

```
Set Integer x = 12
display (x >> 1) ^ 5
```

(A) 1
(B) 3
(C) 5
(D) 7

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** 12>>1=6; 6 XOR 5 = 3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Bit Test with Shift

Q28. What will be the output of the following pseudocode?

```
Set Integer x = 40
display (x >> 3) & 1
```

(A) 0
(B) 1
(C) 2
(D) 5

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** 40>>3=5; 5 AND 1 = 1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Shifted Difference

Q29. What will be the output of the following pseudocode?

```
Set Integer a = 31
display (a << 1) - (a >> 2)
```

(A) 54
(B) 55
(C) 56
(D) 57

Correct Answer: Option (B) : 55

■ **Step-by-Step Logic:** 31<<1=62; 31>>2=7; 62–7=55.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Arithmetic

Q30. What will be the output of the following pseudocode?

```
Integer calc(Integer a, Integer b)
a = a + b
b = a - b
return a * b
End function
display calc(7, 3)
```

(A) 60
(B) 70
(C) 80
(D) 90

Correct Answer: Option (B) : 70

■ **Step-by-Step Logic:** a becomes 10; b becomes 7; return 10×7=70.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Shift

Q31. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return (n << 2) + (n >> 1)
End function
display f(10)
```

(A) 40
(B) 45
(C) 50
(D) 55

Correct Answer: Option (B) : 45

■ **Step-by-Step Logic:** 10<<2=40; 10>>1=5; total=45.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with XOR

Q32. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
return (a ^ b) * 2
End function
display f(12, 5)
```

(A) 18
(B) 26
(C) 30
(D) 34

Correct Answer: Option (A) : 18

■ **Step-by-Step Logic:** 12 XOR 5 = 9; 9×2=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Modulo

Q33. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return (n * n + 3) % 10
End function
display f(7)
```

(A) 2
(B) 5
(C) 8
(D) 9

Correct Answer: Option (A) : 2

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (A) (2).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Boolean

Q34. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
if ((a & 1) == 1 and (b & 1) == 0)
return a + b
else
return a - b
end-if
End function
display f(9, 6)
```

(A) 3
(B) 9
(C) 15
(D) 54

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** 9 is odd and 6 is even, so return 9+6=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Shifted Return

Q35. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer a = n << 1
Set Integer b = n >> 1
return a - b
End function
display f(13)
```

(A) 13
(B) 19
(C) 20
(D) 21

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** 13<<1=26; 13>>1=6; difference=20. Infosys Medium Pseudocode Practice • Page 43

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Multiple Updates

Q36. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
a = a * 2
b = b + a
a = a - b
return a + b
End function
display f(4, 3)
```

(A) 7
(B) 8
(C) 9
(D) 10

Correct Answer: Option (B) : 8

■ **Step-by-Step Logic:** a=8; b=11; a=-3; return -3+11=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Condition + Shift

Q37. What will be the output of the following pseudocode?

```
Set Integer x = 19
if ((x & 1) == 1)
x = x << 1
x = x + 3
else
x = x >> 1
end-if
display x
```

(A) 19
(B) 38
(C) 41
(D) 44

Correct Answer: Option (C) : 41

■ **Step-by-Step Logic:** 19 is odd; 19<<1=38; +3=41.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Function + Logical

Q38. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
if (a > 10 and b < 10)
return (a << 1) + b
else
return a + (b << 1)
end-if
End function
display f(12, 7)
```

(A) 24
(B) 26
(C) 31
(D) 38

Correct Answer: Option (C) : 31

■ **Step-by-Step Logic:** 12>10 and 7<10, so (12<<1)+7 = 24+7=31.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Function + Modulo + Shift

Q39. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return ((n % 5) << 2) + (n >> 3)
End function
display f(26)
```

(A) 5
(B) 7
(C) 9
(D) 11

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** 26%5=1; 1<<2=4; 26>>3=3; total=7.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q40. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code? Integer division is used.
Integer
a, b
Set
a = 17, b = 5
Print
(a / b) + (a % b)
```

(A) 3
(B) 4
(C) 5
(D) 7

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** Integer division: 17 / 5 = 3. Remainder: 17 % 5 = 2. Printed value = 3 + 2 = 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q41. What will be the output of the following pseudocode?

```
What is printed after the following Pseudo
Code runs?
Integer
a, b
Set
a = 12, b = 8
a
= a + b
b
= a - b
a
= a - b
Print
a, b
```

(A) 12 8
(B) 8 12
(C) 20 8
(D) 20 12

Correct Answer: Option (B) : 8 12

■ **Step-by-Step Logic:** This swaps the two values without a third variable. a = 12+8 = 20, b = 20-8 = 12, a = 20-12 = 8. Output: 8 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q42. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code?
Integer
a, b
Set
a = 9, b = 5
Print
a ^ b
```

(A) 4
(B) 12
(C) 13
(D) 14

Correct Answer: Option (B) : 12

■ **Step-by-Step Logic:** 9 in binary is 1001 and 5 is 0101. XOR sets a bit where the bits differ: 1100, which is 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q43. What will be the output of the following pseudocode?

```
What is printed?
Integer
x
Set
x = 10
Print
x & 3
```

(A) 1
(B) 2
(C) 3
(D) 8

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** 10 is 1010 and 3 is 0011. AND is 0010, which is 2.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q44. What will be the output of the following pseudocode?

```
What is printed?
Stack
S
push(S,
1)
push(S,
4)
push(S,
9)
pop(S)
pop(S)
Print
peek(S)
```

(A) 1
(B) 4
(C) 9
(D) Empty

Correct Answer: Option (A) : 1

■ **Step-by-Step Logic:** After three pushes the top is 9. Two pops remove 9 then 4. peek returns 1.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q45. What will be the output of the following pseudocode?

```
[Bitwise]
What is printed?
Integer
a
Set
a = 6
Print
a | 3
```

(A) 7
(B) 5
(C) 2
(D) 18

Correct Answer: Option (A) : 7

■ **Step-by-Step Logic:** 6 is 0110, 3 is 0011, OR is 0111 = 7.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q46. What will be the output of the following pseudocode?

```
[Queue]
What is printed?
Queue
Q
enqueue(Q,
11)
enqueue(Q,
22)
enqueue(Q,
33)
dequeue(Q)
dequeue(Q)
Print
front(Q)
```

(A) 11
(B) 22
(C) 33
(D) Empty

Correct Answer: Option (C) : 33

■ **Step-by-Step Logic:** Queue starts [11,22,33]. Two dequeues remove 11 then 22. Front is 33.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q47. What will be the output of the following pseudocode?

```
[Data
Structures] Queue Q receives 5, then 15, then 25. One
dequeue
runs. What is front(Q)?
Queue
Q
enqueue(Q,
5)
enqueue(Q,
15)
enqueue(Q,
25)
dequeue(Q)
Print
front(Q)
```

(A) 5
(B) 15
(C) 25
(D) Empty

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** FIFO: 5 leaves. Front is 15.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q48. What will be the output of the following pseudocode?

```
[Data
Structures] After the operations, what does peek return?
Stack
S
push(S,
10)
push(S,
20)
push(S,
30)
pop(S)
push(S,
40)
Print
peek(S)
```

(A) 10
(B) 20
(C) 40
(D) 30

Correct Answer: Option (C) : 40

■ **Step-by-Step Logic:** Stack after pops/pushes: [10, 20, 40]. peek = 40.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q49. What will be the output of the following pseudocode?

```
[Bitwise]
What is printed?
Integer
a, b
Set
a = 13, b = 6
Print
a & b
```

(A) 4
(B) 5
(C) 7
(D) 11

Correct Answer: Option (A) : 4

■ **Step-by-Step Logic:** 13 is 1101, 6 is 0110, AND is 0100 = 4.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q50. What will be the output of the following pseudocode?

```
What
is printed?
Stack
S
push(S,
3)
push(S,
6)
push(S,
9)
pop(S)
push(S,
12)
Print
peek(S)
```

(A) 3
(B) 6
(C) 12
(D) 9

Correct Answer: Option (C) : 12

■ **Step-by-Step Logic:** After pop, 9 is gone. 12 is pushed. peek returns 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q51. What will be the output of the following pseudocode?

```
What
is printed?
Integer
x
Set
x = 11
Print
x | 4
```

(A) 15
(B) 7
(C) 8
(D) 44

Correct Answer: Option (A) : 15

■ **Step-by-Step Logic:** 11 is 1011, 4 is 0100, OR is 1111 = 15.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q52. What will be the output of the following pseudocode?

```
What
is printed?
Queue
Q
enqueue(Q,
7)
enqueue(Q,
14)
enqueue(Q,
21)
dequeue(Q)
enqueue(Q,
28)
Print
front(Q)
```

(A) 7
(B) 14
(C) 21
(D) 28

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** 7 is removed first. The new front is 14.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q53. What will be the output of the following pseudocode?

```
What
is printed?
Integer
z
Set
z = 14
Print
z & 7
```

(A) 6
(B) 7
(C) 8
(D) 21

Correct Answer: Option (A) : 6

■ **Step-by-Step Logic:** 14 is 1110, 7 is 0111, AND is 0110 = 6.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q54. What will be the output of the following pseudocode?

```
What
is printed?
Queue
Q
enqueue(Q,
4)
enqueue(Q,
8)
enqueue(Q,
12)
dequeue(Q)
Print
front(Q)
```

(A) 4
(B) 8
(C) 12
(D) Empty

Correct Answer: Option (B) : 8

■ **Step-by-Step Logic:** 4 leaves first. Front is 8.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q55. What will be the output of the following pseudocode?

```
What
is printed?
Stack
S
push(S,
2)
push(S,
5)
push(S,
8)
pop(S)
push(S,
11)
Print
peek(S)
```

(A) 2
(B) 5
(C) 11
(D) 8

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** Pop removes 8. Push 11. peek returns 11.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q56. What will be the output of the following pseudocode?

```
What
is printed?
Stack
S
push(S,
6)
push(S,
9)
push(S,
15)
pop(S)
push(S,
21)
Print
peek(S)
```

(A) 6
(B) 9
(C) 21
(D) 15

Correct Answer: Option (C) : 21

■ **Step-by-Step Logic:** Pop removes 15. Push 21. peek returns 21.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q57. What will be the output of the following pseudocode?

```
What
is printed?
Integer
k
Set
k = 21
Print
k >> 2
```

(A) 5
(B) 6
(C) 10
(D) 84

Correct Answer: Option (A) : 5

■ **Step-by-Step Logic:** 21 is 10101. Right shift 2 gives 101 = 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q58. What will be the output of the following pseudocode?

```
What
is printed?
Queue
Q
enqueue(Q,
3)
enqueue(Q,
9)
enqueue(Q,
15)
dequeue(Q)
Print
front(Q)
```

(A) 3
(B) 9
(C) 15
(D) Empty

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** 3 leaves. Front is 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q59. What will be the output of the following pseudocode?

```
What will be the
output of the following pseudo code
Integer
i, j, k, d
Set
j = 6, i = 8, k = 2, d = 1
if
((j ^ i) mod k equals 0) then
d = d + 1
else
d = d - 1
end
if
Print
d
```

(A) 1
(B) 3
(C) 0
(D) 2

Correct Answer: Option (D) : 2

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (2).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q60. What will be the output of the following pseudocode?

```
1.     Where are placed the list of processes that are
prepared to be executed and waiting?
```

(A) Job queue
(B) Ready queue
(C) Execution queue
(D) Process queue

Correct Answer: Option (B) : Ready queue

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (Ready queue).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q61. What will be the output of the following pseudocode?

```
1.    Who among the following can block the running
process?
```

(A) Fork
(B) Read
(C) Down
(D) All of these

Correct Answer: Option (D) : All of these

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (All of these).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q62. What will be the output of the following pseudocode?

```
1.    Which of the following does not interrupt the
running process?
```

(A) Timer interrupt
(B) Device
(C) Power failure
(D) Scheduler process

Correct Answer: Option (D) : Scheduler process

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (Scheduler process).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q63. What will be the output of the following pseudocode?

```
1.    The \u0021 article referred to as a
```

(A) Unicode escape sequence
(B) Octal escape
(C) Hexadecimal
(D) Line feed

Correct Answer: Option (A) : Unicode escape sequence

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (Unicode escape sequence).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q64. What will be the output of the following pseudocode?

```
1.    Which of the following is a valid long literal?
```

(A) ABH8097
(B) L990023
(C) 904423
(D) 0xnf029L

Correct Answer: Option (D) : 0xnf029L

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (0xnf029L).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q65. What will be the output of the following pseudocode?

```
1.    What does the expression float a = 35 / 0 return?
```

(A) 0
(B) Not a Number
(C) Infinity
(D) Run time exception

Correct Answer: Option (C) : Infinity

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (Infinity).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q66. What will be the output of the following pseudocode?

```
1.    Which option is false about the final keyword?
a)   A) A final method cannot be overridden in its
subclasses.
b)   B) A final class cannot be extended.
c)   C) A final class cannot extend other classes.
d)   D) A final method can be inherited.
```

(A) A&B
(B) B&C
(C) C&D
(D) None of these

Correct Answer: Option (C) : C&D

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (C) (C&D).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q67. What will be the output of the following pseudocode?

```
1.    Construct post order Transversal for the
constructed tree with the given In-Order and Preorder
Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

(A) 9 10 7 8 6
(B) 9 7 6 10 8
(C) 8 9 10 7 6
(D) 9 8 10 7 6

Correct Answer: Option (A) : 9 10 7 8 6

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (9 10 7 8 6).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q68. What will be the output of the following pseudocode?

```
1.    Which of the following declarations is invalid in
C language?
```

(A) char
(B) char str[] = “campus monk is the best platform for learn";
(C) char str[20] = “campus monk is the best platform for learn";
(D) char[] str = “campus monk is the best platform for learn";

Correct Answer: Option (D) : char[] str = “campus monk is the best platform for learn";

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (char[] str = “campus monk is the best platform for learn";).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q69. What will be the output of the following pseudocode?

```
1.     In an empty BST, insert the following elements in
a similar sequence. Which element will be at lowest
level?

105     95    122    99    97   120
```

(A) 95
(B) 97
(C) 99
(D) 120

Correct Answer: Option (B) : 97

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (97).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q70. What will be the output of the following pseudocode?

```
1.     Which of the following sequence can be obtained in
the output (in same order) using a stack assuming that
the input is in the sequence 1,2,3,4,5 order?
```

(A) a) 3,4,5,1,2
(B) a) 3,4,5,2,1
(C) a) 1,5,2,3,4
(D) a) 5,4,3,1,2

Correct Answer: Option (B) : a) 3,4,5,2,1

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (a) 3,4,5,2,1).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q71. What will be the output of the following pseudocode?

```
1.     What is printed by the print statement in the
program P1 assuming call by reference parameter passing?
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z

}
```

(A) 10, 3
(B) 31, 3
(C) 27, 7
(D) None of these

Correct Answer: Option (B) : 31, 3

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (31, 3).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q72. What will be the output of the following pseudocode?

```
Integer a, b, c Set a = 2, b = 4, c = 7 b = 7 + a if (c >
b) b = b + b end if b = 6 + a if ((a + b) < (b - a) || c <
a) b = (c + 4) + c else a = c + b end if c = (c + 9) ^ a
Print a + b + c
```

(A) 71
(B) 56
(C) 47
(D) 54

Correct Answer: Option (D) : 54

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (54).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q73. What will be the output of the following pseudocode?

```
Int p, q, r Set p = 6, q = 4, r = 7 p = (r + 5) + p If ((q
& r & p) < (p & 6)) { r = 3 ^ r q = q & q } Print p + q +
r
```

(A) 44
(B) 41
(C) 11
(D) 26

Correct Answer: Option (D) : 26

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (26).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q74. What will be the output of the following pseudocode?

```
What is the output of this code? Int p, q, r; Set p = 0,
q = 7, r = 9; If ((q + p) < (p + q) && (p < r)) { p = 11 +
q; } Print p + q + r;
```

(A) 12
(B) 16
(C) 27
(D) 10

Correct Answer: Option (B) : 16

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (16).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q75. What will be the output of this pseudocode? a = 7, b = 2, c = 8 If ((b ^ a) < a && (2 ^ 7) < c) a = b + c Else c = (4 + 12) + a b = 11 + a End if Print (a + b + c)

(A) 19
(B) 21
(C) 20
(D) 22

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (20).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q76. What will be the output of the following pseudocode?

```
Find the output of the code snippet. Integer x, y, z Set x = 1, y = 8, z = 2 If (x ^ y && x ^ y && x ^ y ^ z) x = 1 End if Print x + y + z
```

(A) 11
(B) 13
(C) 16
(D) 9

Correct Answer: Option (A) : 11

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (11).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q77. What will be the output of the following pseudocode?

```
Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x - 2 y = x Else x = z y = y ^ 2 End if Print x + y + z
```

(A) 2
(B) 11
(C) 3
(D) 0

Correct Answer: Option (C) : 3

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (3).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q78. What will be the output of the following pseudocode?

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x) return 0; }
```

(A) 0
(B) 1
(C) 5
(D) Compilation Error

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q79. What will be the output of the following pseudocode?

```
Q4.) Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x -2 y = x Else x = z y = y ^ 2 End if Print x + y + z
```

(A) 2
(B) 11
(C) 0
(D) 3

Correct Answer: Option (D) : 3

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (3).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q80. What will be the output of the following pseudocode?

```
Q5. What is the output int main() { int x = 10; int *y, **z; y = &x; z = &y; **z=7; print(x,*y,**z) return 0; }
```

(A) 10 10 10
(B) 10 7 7
(C) 7 7 7
(D) 10 10 7

Correct Answer: Option (C) : 7 7 7

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (7 7 7).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q81. Q8. What will be the output of the following pseudocode? int a , b ,c Set a=0,b=1,c=2 If (b^c || a&b || a>>1) c=9 a =b+c else c =1 a=a+c End if Print a+b+c

(A) 20

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q82. What will be the output of the following pseudocode?

```
What is the output of following code: Int main(){ Int n=(1<<2+3<<4) Print(n); Return 0; }
```

(A) 112
(B) 512
(C) 52
(D) 0

Correct Answer: Option (B) : 512

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (512).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q83. What will be the output of the following pseudocode?

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }
```

(A) 0
(B) 1
(C) 5
(D) Compilation Error

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q84. What will be the output of the following pseudocode?

```
Integer p, q, r Set p = 4, q = 2, r = 5 p = (q + p) + p
if ((q + p) < (r - q) && 3 > p) then p = (q + r) + p if
(((p + r) + (r + q)) <= (p + r + q)) then r = q + p else
r = (p + r) + p end if p = (9 & 6) + q else q = r & r if
((q - 5) < (5 - q)) then r = (6 + 6) & q q = (q + 7) + r
end if p = (1 ^ 11) + q end if Print p + q + r
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q85. What will be the output of the following pseudocode?

```
Integer a, b, c Set a = 2, b = 4, c = 7 b = 7 + a if (c >
b) b = b + b end if b = 6 + a if ((a + b) < (b - a) || c <
a) b = (c + 4) + c else a = c + b end if c = (c + 9) ^ a
Print a + b + c
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q86. What will be the output of the following pseudocode?

```
Integer a, b, c Set a = 1, b = 6, c = 7 b = (a & b) + b
for (each c from 3 to 5) { b = (a + a) ^ a b = (12 + 5) +
a } a = (a ^ 9) + a Print a + b
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q87. What will be the output of the following pseudocode?

```
A stack is used (operands are evaluated left to right).
What is printed?
push(3); push(5); push(7);
x = pop();
push(x * 2);
push(pop() + pop());
print(pop() - pop())
```

(A) 14
(B) 16
(C) 19
(D) 22

Correct Answer: Option (A) : 14

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (14).

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q88. What will be the output of the following pseudocode?

```
What is printed?
```

(A) 40
(B) 46
(C) 52
(D) 44

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q89. What will be the output of the following pseudocode?

```
What is printed?
```

(A) 37
(B) 41
(C) 45
(D) 39

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q90. What will be the output of the following pseudocode?

```
What is printed?
```

(A) 130
(B) 124
(C) 136
(D) 118

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q91. What will be the output of the following pseudocode?

```
What is printed? (^=XOR, |=OR, &=AND, >>=right shift)
a = 13; b = 25;
c = (a ^ b) & (a | b);
d = c >> 2;
print(c + d)
```

(A) 25
(B) 29
(C) 20
(D) 33

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 5]

Topic: *Infosys SE College Mock Test*

Q92. What will be the output of the following pseudocode?

```
This stack trace evaluates a postfix-style expression.
What is printed?
push(6); push(2); push(3);
c = pop(); b = pop(); push(b + c);
d = pop(); e = pop(); push(e - d);
push(3); push(8); push(2);
f = pop(); g = pop(); push(g / f);
h = pop(); i = pop(); push(i + h);
j = pop(); k = pop(); push(k * j);
print(pop())
```

(A) 7

(B) 10

(C) 14

(D) 4

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches .

[College ITM Mock Paper 6]

Topic: *Infosys SE College Mock Test*

Q93. What will be the output of the following pseudocode?

```
What is printed? (^=XOR, >>=right shift, &=AND)
n = 13
g = n ^ (n >> 1)
print(g + (g & 5))
```

(A) 12

(B) 11

(C) 16

(D) 14

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .